

Incident Response & Log Analysis Report

Project Title: Log Analysis & Alerting Tool for a Retail Chain

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Tool Used: Windows Event Viewer (`eventvwr.msc`)

1. Project Overview & Objective

In a retail chain environment, Point-of-Sale (POS) terminals, back-office servers, and inventory management systems generate large volumes of log data continuously. Manual log review is inefficient and prone to missing critical security incidents.

The objective of this module is to implement targeted log filtering and detection rules using **Windows Event Viewer** to monitor suspicious activities, prevent unauthorized access, ensure POS operational continuity, and establish standardized incident response procedures.

2. Configured Detection & Alert Rules

To monitor high-risk events across the retail infrastructure, five custom detection rules (Custom Views) were defined based on specific Windows Security and System Event IDs:

Rule 1: Failed Login Attempts (Brute Force Detection)

Event ID: `4625`

Log Source: Security

Severity: Medium

Description: Triggers when an invalid username or password is submitted during a logon attempt on a POS terminal or management server. Multiple consecutive occurrences indicate a potential brute-force attack.

Rule 2: Security Log Cleared (Log Tampering)

Event ID: `1102`

Log Source: Security

Severity: High / Critical

Description: Triggers when the Windows Audit Security Log is explicitly cleared. Attackers often attempt to wipe audit logs to conceal malicious activity, privilege escalation, or unauthorized access.

Rule 3: User Account Locked Out

Event ID: `4740`

Log Source: Security

Severity: Medium

Description: Triggers when a user account is automatically locked out after exceeding the maximum allowed invalid logon attempts.

Rule 4: New User Account Created

Event ID: `4720`

Log Source: Security

Severity: High

Description: Triggers whenever a new user account is created on the local system or domain controller. Unauthorized account creation can indicate persistence mechanisms established by an adversary.

Rule 5: System Restart / Shutdown (POS Downtime Event)

Event ID: `1074`

Log Source: System

Severity: Low / Medium

Description: Triggers when an application or user initiates a system shutdown or reboot. Tracking restarts ensures system availability and identifies unauthorized reboots on active checkout terminals.

3. Incident Response Matrix & Action Steps

When an alert is triggered in the monitoring system, the following standard operating procedures (SOPs) must be executed:

Rule Name	Event ID	Severity	Threat / Risk	Incident Response Steps
1.Failed Logins	4625	Medium	Unpermitted access attempt / Password guessing	1. Identify the targeted Account Name and Source IP Address.

				2. Check if attempts exceed the threshold (> 5 within 5 minutes). 3. Temporarily disable the account if automated brute-forcing is suspected. 4. Notify the store manager and verify if legitimate staff made a credential mistake.
2.Log Cleared	1102	High	Antforensic activity / Hiding malicious traces	1. Immediately isolate the affected host machine from the retail network. 2. Identify the Subject Security ID/User who performed the clear action. 3. Terminate active sessions associated with that user account. 4. Escalate to the Tier-2 Security Incident Response team for disk/memory forensics.
3.Account Lockout	4740	Medium	Unauthorized access attempt or forgotten credentials	1. Cross-check the locked username with store shift schedules.

				2. Contact the employee via an out-of-band channel (phone/email) to verify identity. 3. Inspect source computer logs for unauthorized automated scripts. 4. Unlock the account and prompt for a password reset if identity is confirmed.
4. New User Created	4720	High	Unauthorized privilege expansion / Backdoor user	1. Identify the creator account (SubjectUserName) and newly created username (TargetUserName). 2. Verify against HR and IT change-management ticket approvals. 3. If unauthorized, immediately disable and delete the newly created user account. 4. Revoke privileges of the creator account pending investigation.
5. System Restart	1074	Medium	POS terminal downtime / Unscheduled interruption	1. Determine whether the shutdown was user-initiated, planned maintenance, or an error.

				2. Confirm operational status of the POS register/checkout application. 3. If unscheduled during active store operating hours, inspect event details for process name and user ID.
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4. Verification & Testing

1. Custom Views Setup: The 5 rules were created under `Event Viewer -> Custom Views` using specific XML/Log filters.

2. Log Review Verification: Filters were queried against actual system logs to ensure clean event separation without administrative overhead or false positives.

3. Evidence Capture: Screenshots of configured Custom Views and event details were archived for project audit and documentation.

